

IMPLICITMEMBENCH: Measuring Unconscious Behavioral Adaptation in Large Language Models

IMPLICITMEMBENCH: 测量大型语言模型中的无意识行为适应

Chonghan Qin¹, Xiachong Feng^{1*}, Weitao Ma², Xiaocheng Feng², Lingpeng Kong¹

秦崇翰¹, 冯霞冲^{1*}, 马伟涛², 冯晓城², 宋灵鹏¹

¹ The University of Hong Kong

¹ 香港大学

² Harbin Institute of Technology

² 哈尔滨工业大学

qinch@connect.hku.hk, fengxc@hku.hk

qinch@connect.hku.hk, fengxc@hku.hk

Abstract

摘要

Existing memory benchmarks for LLM agents evaluate explicit recall of facts, yet overlook implicit memory where experience becomes automated behavior without conscious retrieval. This gap is critical: effective assistants must automatically apply learned procedures or avoid failed actions without explicit reminders. We introduce IMPLICITMEMBENCH, the first systematic benchmark evaluating implicit memory through three cognitively grounded constructs drawn from standard cognitive-science accounts of non-declarative memory: Procedural Memory (one-shot skill acquisition after interference), Priming (theme-driven bias via paired experimental/control instances), and Classical Conditioning (Conditioned Stimulus–Unconditioned Stimulus (CS–US) associations shaping first decisions). Our 300-item suite employs a unified Learning/Priming-Interfere-Test protocol with first-attempt scoring. Evaluation of 17 models reveals severe limitations: no model exceeds 66% overall, with top performers DeepSeek-R1 (65.3%), Qwen3-32B (64.1%), and GPT-5 (63.0%) far below human baselines. Analysis uncovers dramatic asymmetries (inhibition 17.6% vs. preference 75.0%) and universal bottlenecks requiring architectural innovations beyond parameter scaling. IMPLICITMEMBENCH reframes evaluation from “what agents recall” to “what they automatically enact”¹.

现有的 LLM 代理记忆基准测试评估了显性回忆事实的能力，却忽略了隐性记忆，即经验转化为无需意识检索的自动化行为。这一差距至关重要：有效的助手必须在没有明确提示的情况下自动应用所学程序或避免失败行为。我们引入了 IMPLICITMEMBENCH，这是首个通过三个基于标准认知科学对非陈述性记忆的理论构建的系统性基准测试，来评估隐性记忆的基准测试：程序性记忆（在干扰后一次性技能获取）、启动效应（通过配对实验/对照实例驱动主题偏倚）以及经典条件反射（条件刺激-非条件刺激 (CS-US) 关联塑造首次决策）。我们的 300 项测试套件采用统一的学习/启动-干扰-测试协议，并采用首次尝试评分。对 17 个模型的评估揭示了严重的局限性：没有任何模型的总体得分超过 66%，表现最好的模型 DeepSeek-R1 (65.3%)、Qwen3-32B (64.1%) 和 GPT-5 (63.0%) 远低于人类基准。分析揭示了显著的不对称性（抑制 17.6% vs. 偏好 75.0%）以及需要超越参数扩展的架构创新的普遍瓶颈。IMPLICITMEMBENCH 将评估重点从“代理能回忆什么”转变为“代理能自动执行什么”¹。

1 Introduction

1 引言

Large language models (LLMs) have demonstrated remarkable capabilities across diverse domains (Zhao et al., 2023; Wu et al., 2024), maturing into assistants for daily productivity and specialized workflows (DeepSeek-AI et al., 2025; Zeng et al., 2025; OpenAI, 2025; Anthropic, 2025; Qwen et al., 2024; Yang et al., 2025). To serve users

大型语言模型 (LLMs) 在多个领域展现了显著的能力 (Zhao 等, 2023; Wu 等, 2024), 逐渐发展成为日常生产力和专业工作流程的助手 (DeepSeek-AI 等, 2025; Zeng 等, 2025; OpenAI, 2025; Anthropic, 2025; Qwen 等, 2024; Yang 等, 2025)。为了服务用户

reliably, these systems require memory: the ability to accumulate and leverage experience across interactions. Recent work establishes this need (He et al., 2024; Zhang et al., 2025a,b), with production systems deploying memory features (OpenAI, 2024). Benchmarks such as LoCoMo, LongMemEval, MADial-Bench, MemBench, MemoryAgentBench, MEMTRACK, and GoodAI LTM now evaluate multi-session QA, retrieval, dialogue, state tracking, and conversational integration under explicit, actively triggered settings spanning contexts from 400 tokens to 1.5M tokens, or variable long-horizon environments (Maharana et al., 2024; Wu et al., 2025; He et al., 2025; Tan et al., 2025; Hu et al., 2026; Deshpande et al., 2025; Castillo-Bolado et al., 2024).

可靠地, 这些系统需要记忆: 在交互中积累并利用经验的能力。近期的研究证实了这一需求 (He 等, 2024; Zhang 等, 2025a,b), 并且生产系统已经部署了记忆功能 (OpenAI, 2024)。诸如 LoCoMo、LongMemEval、MADial-Bench、MemBench、MemoryAgentBench、MEMTRACK 和 GoodAI LTM 等基准测试, 现在在显式的、主动触发的设置下评估多会话问答、检索、对话、状态跟踪和对话整合, 覆盖从 400 个 token 到 1.5M 个 token 的上下文, 或可变的长视野环境 (Maharana 等, 2024; Wu 等, 2025; He 等, 2025; Tan 等, 2025; Hu 等, 2026; Deshpande 等, 2025; Castillo-Bolado 等, 2024)。

Despite this progress, existing benchmarks predominantly evaluate explicit memory: conscious retrieval of factual information. Table 1 shows prior work uniformly adopts query-response protocols where models are explicitly prompted to recall facts, limiting evaluation to deliberate retrieval rather than unconscious behavioral adaptation. Yet critical failures arise from missing implicit memory, experience that becomes automated behavior rather than explicit recollection. Effective assistants should automatically apply learned procedures after distractions or avoid repeatedly-failed tools without explicit reminders. Existing benchmarks fail to diagnose this because they (i) employ QA formats explicitly cueing target information, (ii) stress storage capacity over first-attempt triggers after interference, and (iii) utilize costly pipelines hindering reproducibility.

尽管取得了这些进展, 现有的基准测试主要评估显性记忆: 有意识地检索事实性信息。表 1 显示, 先前的工作普遍采用查询-响应协议, 其中模型被明确提示回忆事实, 从而将评估限制在有意识的检索, 而非无意识的行为适应。然而, 关键的失败源于缺乏隐性记忆, 即经验自动转化为行为, 而非有意识的回忆。有效的助手应在分心后自动应用所学的程序, 或在没有明确提示的情况下避免重复失败的工具。现有的基准测试未能诊断出这些问题, 因为它们 (i) 采用明确提示目标信息的问答格式, (ii) 强调干扰后存储容量而非首次尝试的触发, 以及 (iii) 使用成本高昂的流程阻碍了可重复性。

We introduce IMPLICITMEMBENCH, a cognitively grounded benchmark evaluating implicit memory in LLM agents. We operationalize three constructs: Procedural Memory assesses one-shot skill acquisition persisting after interference, Priming measures theme-driven biases via paired experimental/control instances, and Classical Condi-

我们介绍了 IMPLICITMEMBENCH, 这是一个基于认知的基准测试, 用于评估大型语言模型代理中的隐性记忆。我们对三个构念进行了操作化定义: 程序性记忆评估在干扰之后仍能持续的单次技能获取, 启动效应通过配对的实验/对照实例测量主题驱动的偏差, 以及经典条件反射。

Benchmark	Memory Type	Evaluation Trigger	Context Scale (tokens)	Evaluation Size	Task Focus
LoCoMo (Maharana et al., 2024)	Explicit	Active(Explicit query)	9K	300 conversations	Multi-session QA
LongMemEval (Wu et al., 2025)	Explicit	Active(Explicit query)	115K–1.5M	500 questions	Information retrieval
MADial-Bench (He et al., 2025)	Explicit	Active(Emotion cued)	400	160 dialogues	Emotional dialogue
MemBench (Tan et al., 2025)	Explicit	Active(Explicit query)	1K–100K	53,000 QAs	Reflective memory during observation
MemoryAgentBench (Hu et al., 2026)	Explicit	Active(Explicit query)	100K–1.4M	14 datasets / 2,071 QAs	4 Competencies (Retrieval, Test-time Learning, Long Range Understanding, Selective Forgetting)
MEMTRACK (Deshpande et al., 2025)	Explicit	Active(Explicit task)	Variable	210 instances	Multi-platform State Tracking
GoodAI LTM (Castillo-Bolado et al., 2024)	Explicit	Active(Explicit query)	2K–500K	Variable	Conversational / Info Integration
IMPLICITMEMBENCH (Ours)	Implicit	Passive(Scenario)	500	300 instances	Unconscious adaptation

基准测试	内存类型	评估触发器	上下文规模 (tokens)	评估规模	任务聚焦
LoCoMo (Maharana 等, 2024)	显性	主动 (显式查询)	9K	300 次对话	多会话问答
LongMemEval (Wu 等, 2025)	显性	主动 (显式查询)	115K–1.5M	500 个问题	信息检索
MADial-Bench (He et al., 2025)	显性	主动 (情绪提示)	400	160 对话	情感对话
MemBench (Tan et al., 2025)	显性	主动 (显式查询)	1K–100K	53,000 个问答	观察期间的反射记忆

MemoryAgentBench (Hu et al., 2026)	显性	主动（显式查询）	10 万–140 万	14 个数据集 / 2,071 个问答	4 种能力（检索、测试时学习、长距离理解、选择性遗忘）
MEMTRACK (Deshpande 等, 2025)	显性	Active(显式任务)	变量	210 个实例	多平台状态追踪
GoodAI LTM (Castillo-Bolado 等, 2024)	显性	主动（显式查询）	2K–500K	变量	对话 / 信息整合
IMPLICITMEMBENCH (Ours)	隐性	被动（场景）	500	300 个实例	无意识适应

Table 1: Comparison of IMPLICITMEMBENCH with existing memory benchmarks. Most prior benchmarks target explicit memory through active retrieval triggers, whereas IMPLICITMEMBENCH evaluates implicit memory through passive, scenario-driven behavioral adaptation.

表 1: IMPLICITMEMBENCH 与现有内存基准的比较。大多数先前的基准测试通过主动检索触发器针对显性记忆，而 IMPLICITMEMBENCH 通过被动的、基于场景的行为适应来评估隐性记忆。

tioning evaluates whether Conditioned Stimulus–Unconditioned Stimulus (CS–US) exposure shapes first decisions. Our paradigm selection is guided by the classical taxonomy of non-declarative memory (Squire, 2004). We focus on three mechanisms that are especially relevant to LLM agents: procedural memory for internalizing new routines, priming for context-driven adaptation without explicit instruction, and classical conditioning for forming automatic protective responses from experience. We exclude non-associative learning in this version because it is less directly aligned with the high-level semantic decision-making emphasized in current agentic systems. This grounding lets us map established cognitive constructs to text-based agent evaluation through functional isomorphism rather than surface analogy. All 300 items follow a unified Learning/Priming–Interfere–Test protocol with first-attempt scoring isolating automatized behavior from explicit recall. Figure 1 illustrates our framework: procedural tasks use rule-based validators, priming employs LLM judges comparing experimental versus control conditions, and classical conditioning tracks binary first-attempt avoidance. This design enables lightweight, reproducible evaluation revealing that implicit memory formation poses profound challenges with no model achieving human-like automaticity.

评估是否条件刺激–无条件刺激（CS–US）暴露塑造了最初的决策。我们的范式选择受到经典非陈述性记忆分类（Squire, 2004）的指导。我们关注三种特别适用于大型语言模型代理的机制：用于内化新常规的程序性记忆、无需明确指导即可进行情境驱动适应的启动效应，以及通过经验形成自动保护反应的经典条件反射。在此版本中，我们排除了非关联学习，因为它与当前代理系统中强调的高级语义决策的关联性较弱。这种基础使我们能够通过功能同构而非表面类比，将已有的认知构念映射到基于文本的代理评估中。所有 300 个项目均遵循统一的学习/启动–干扰–测试协议，首次尝试评分能够将自动化行为与显性回忆区分开来。图 1 展示了我们的框架：程序性任务使用基于规则的验证器，启动效应采用大型语言模型评判者，比较实验条件与对照条件，经典条件反射追踪首次尝试的二元回避行为。这种设计使评估轻量化且可重复，揭示了隐性记忆形成带来了深远的挑战，没有任何模型能够达到类似人类的自动化水平。

We evaluate 17 models spanning proprietary

我们评估了 17 个模型，涵盖专有模型

(GPT-5, Claude-4.1-opus, Gemini-2.5-pro) and open-source systems (DeepSeek-R1, Qwen3-32B, LLaMA-3.3-70B). As Figure 2 shows, results reveal fundamental limitations. First, a severe ceiling effect emerges: top performers remain far below human baselines, with no model exceeding two-thirds overall accuracy. Second, paradigm asymmetry shows dramatic variance: procedural

memory proves most tractable while classical conditioning creates substantial bottlenecks, and priming clusters in a narrow moderate range. Third, capability dissociation reveals that excellence in one paradigm fails to predict success in others, with the strongest procedural learner suffering dramatic drops on classical conditioning. Fine-grained analysis uncovers systematic failures: models struggle profoundly with inhibition versus preference-based learning, and multiple categories remain universally challenging across all architectures.

(GPT-5, Claude-4.1-opus, Gemini-2.5-pro) 和开源系统 (DeepSeek-R1, Qwen3-32B, LLaMA-3.3-70B)。如图 2 所示, 结果揭示了基本的局限性。首先, 出现严重的天花板效应: 表现最好的模型仍远低于人类基准水平, 没有任何模型的总体准确率超过三分之二。其次, 范式不对称性表现出显著的差异: 程序性记忆证明是最容易处理的, 而经典条件反射则造成了显著的瓶颈, 而启动效应则集中在狭窄的中等范围内。第三, 能力解耦表明, 在一个范式中表现出色并不能预测在其他范式中的成功, 最强的程序性学习者在经典条件反射任务中会出现显著的下降。细致分析揭示了系统性失败: 模型在抑制与偏好驱动学习之间存在严重困难, 多个类别在所有架构中都普遍具有挑战性。

Our main contributions are: (1) We present the first benchmark of implicit memory in LLMs, operationalizing procedural learning, priming, and classical conditioning under a unified protocol isolating automatized behavior through first-attempt scoring. (2) We design an evaluation framework combining rule-based validators and LLM judges, enabling reproducible assessment via a compact 300-item suite. (3) We evaluate 17 models revealing critical weaknesses: severe behavioral asymmetries

我们的主要贡献是: (1) 我们提出了首个针对大语言模型中隐式记忆的基准测试, 通过统一的协议, 利用首次尝试评分来隔离自动化行为, 将程序性学习、启动效应和经典条件反射进行操作化定义。(2) 我们设计了一个结合基于规则的验证器和大语言模型评判者的评估框架, 通过一个紧凑的 300 项套件实现可重复的评估。(3) 我们评估了 17 个模型, 揭示了关键的弱点: 严重的行为主观不对称性

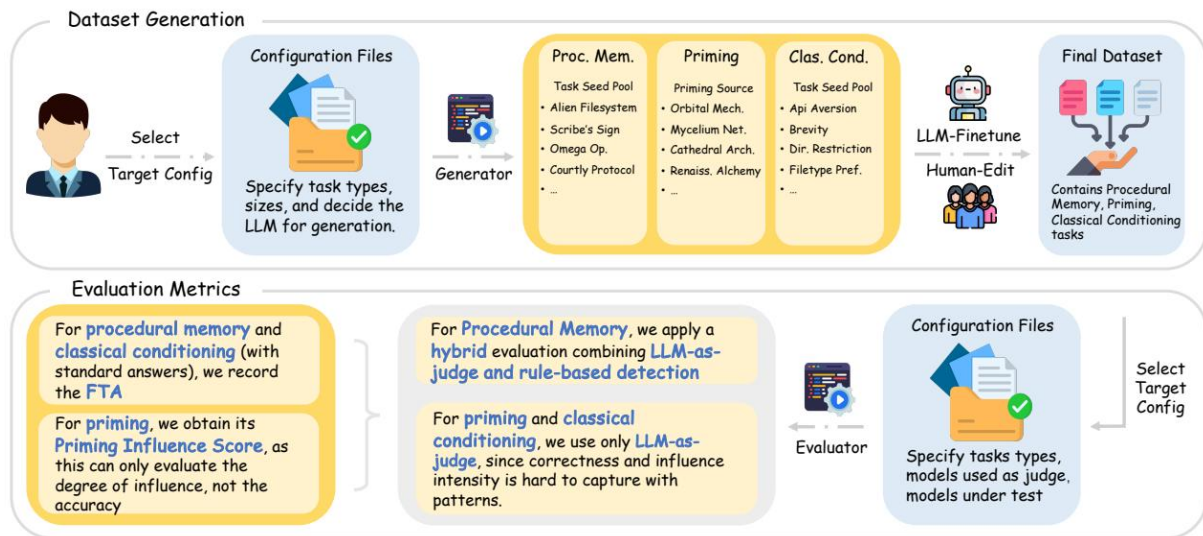


Figure 1: images/ed7c4af8b4a30475318e661c0bbb29d4ef9168d078c5bbfa61e2c86d30fae59e.jpg

Figure 1: Overall framework. (a) Dataset generation pipeline using LLM-generated candidates refined via fine-tuning and human editing. (b) Evaluation metrics: hybrid automatic/human validation for Procedural Memory (FTA), LLM-as-Judge for Priming (Priming Influence Score) and Classical Conditioning (FTA).

图 1: 总体框架。(a) 使用大型语言模型生成的候选数据集生成流程, 通过微调和人工编辑进行优化。(b) 评估指标: 程序性记忆 (FTA) 的混合自动/人工验证, 作为评判者的大型语言模型用于启动 (启动影响分数) 和经典条件反射 (FTA)。

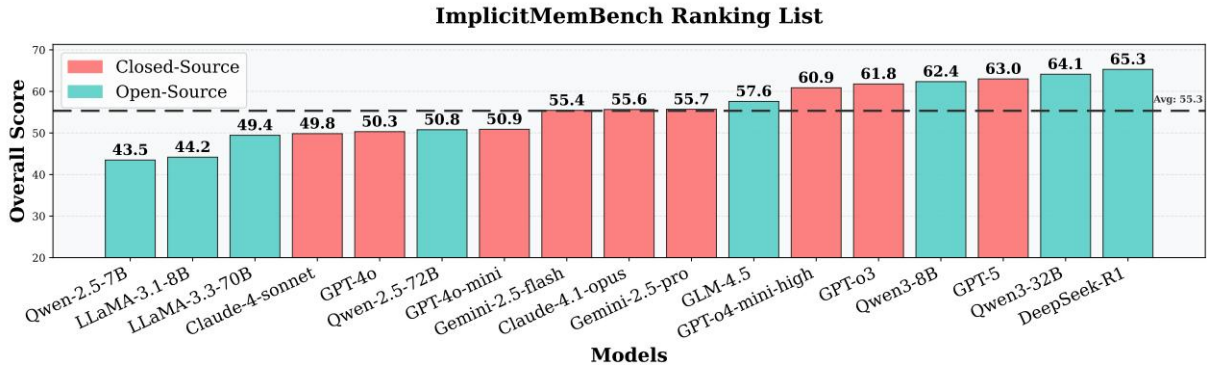


Figure 2: images/f7d31e4b04b333d5cbabad9dab9ac8303a633153d8fd08118d4d68c6bb5fedb6.jpg

Figure 2: Performance ranking of all evaluated models on IMPLICITMEMBENCH. Models are sorted by overall score, with closed-source models shown in red-coral and open-source models in teal. The horizontal dashed line indicates the average performance across all models.

图 2：所有评估模型在 IMPLICITMEMBENCH 上的性能排名。模型按照总体得分排序，闭源模型用红色珊瑚色表示，开源模型用青绿色表示。水平虚线表示所有模型的平均性能。

and universal bottlenecks requiring architectural innovations beyond parameter scaling. (4) We further show that representative memory-augmented agents do not consistently improve performance on IMPLICITMEMBENCH, suggesting that implicit memory cannot be reduced to explicit storage and retrieval alone.

以及需要超越参数扩展的架构创新的通用瓶颈。(4) 我们进一步表明，代表性记忆增强智能体在 IMPLICITMEMBENCH 上的表现并不一致，这表明隐式记忆不能仅通过显式存储和检索来简化。

2 Related Work

2 相关工作

Existing memory benchmarks for LLM agents predominantly evaluate explicit memory through active retrieval triggers. LoCoMo (Maharana et al., 2024) studies multi-session dialogue continuity via QA and event summarization over 300 conversations of roughly 9k tokens each. LongMemEval (Wu et al., 2025) examines long-context memory in 115k–1.5M-token settings across re-

现有的针对大型语言模型代理的记忆基准测试主要通过主动检索触发器来评估显式记忆。LoCoMo (Maharana 等, 2024) 通过在大约每个对话包含 9k 个 token 的 300 个对话中进行问答和事件摘要，研究多会话对话的连贯性。LongMemEval (Wu 等, 2025) 则在 115k 至 1.5M 个 token 的设置下考察长上下文记忆，跨重新

trieval and multi-session reasoning tasks. MADial-Bench (He et al., 2025) focuses on emotion-support dialogue in 160 dialogues. MemBench (Tan et al., 2025) expands evaluation to factual and reflective memory across participation/observation scenarios with contexts ranging from 1K to 100K tokens. More recent benchmarks broaden task coverage while retaining the same explicit framing: MemoryAgentBench (Hu et al., 2026) evaluates four competencies—accurate retrieval, test-time learning, long-range understanding, and selective forgetting—over 14 datasets / 2,071 QAs in 100K–1.4M-token settings; MEMTRACK (Deshpande et al., 2025) studies multi-platform state tracking in variable long-horizon environments; and GoodAI LTM (Castillo-Bolado et al., 2024) emphasizes conversational information integration in 2K–500K-

检索和多会话推理任务。MADial-Bench (He 等, 2025) 专注于 160 个对话中的情感支持对话。MemBench (Tan 等, 2025) 将评估范围扩展到参与/观察场景中的事实性和反思性记忆, 上下文范围从 1K 到 100K 个标记。最近的基准测试在保持相同显式框架的同时扩大了任务覆盖范围: MemoryAgentBench (Hu 等, 2026) 在 100K 至 1.4M 个标记的设置下, 针对 14 个数据集/2,071 个问答, 评估了四项能力——准确检索、测试时学习、长距离理解以及选择性遗忘; MEMTRACK (Deshpande 等, 2025) 研究了在可变长周期环境中的多平台状态跟踪; 而 GoodAI LTM (Castillo-Bolado 等, 2024) 则强调在 2K 至 500K 个标记的范围内进行对话信息整合。

token contexts. As summarized in Table 1, prior work covers increasingly diverse tasks and scales, but still targets explicit memory through active triggers, leaving implicit memory unaddressed. IMPLICITMEMBENCH fills this gap as the first systematic evaluation of implicit memory via procedural learning, priming, and classical conditioning, using efficient ~500-token protocols to assess unconscious behavioral adaptations. Beyond benchmark design, another line of work augments LLM agents with external memory modules or retrieval-based long-term memory systems. These methods primarily target explicit storage and recall. As we show in Section D, however, such mechanisms do not consistently improve performance on IMPLICITMEMBENCH, indicating that implicit memory is not reducible to explicit retrieval alone.

token contexts. 如表 1 所示, 先前的工作涵盖了越来越多的多样化任务和规模, 但仍然通过主动触发来针对显性记忆, 而忽略了隐性记忆。IMPLICITMEMBENCH 填补了这一空白, 作为首个通过程序性学习、启动效应和经典条件反射对隐性记忆进行系统评估的基准测试, 使用高效的 ~500 个标记协议来评估无意识的行为适应。除了基准测试设计, 另一条研究路线是通过外部记忆模块或基于检索的长期记忆系统来增强大型语言模型代理。这些方法主要针对显性存储和检索。然而, 如我们在第 D 节所示, 这些机制在 IMPLICITMEMBENCH 上的表现并不一致, 表明隐性记忆不能仅通过显性检索来实现。

3 IMPLICITMEMBENCH

3 IMPLICITMEMBENCH

3.1 Cognitive Grounding and Paradigm Selection

3.1 认知基础与范式选择

Our benchmark is grounded in the taxonomy of non-declarative memory (Squire, 2004). We focus on procedural memory, priming, and classical conditioning because they capture complementary forms of implicit adaptation that are especially relevant to LLM agents. We do not include non-associative learning in the current version, as it is less directly connected to the semantic, decision-oriented behaviors emphasized in modern agent workflows. Our design principle is functional isomorphism: we translate standard cognitive mechanisms into text-based agentic scenarios while preserving their core causal structure.

我们的基准基于非陈述性记忆的分类法 (Squire, 2004)。我们关注程序性记忆、启动效应和经典条件反射, 因为它们捕捉了与 LLM 代理特别相关的互补形式的隐式适应。在当前版本中, 我们不包括非关联学习, 因为它与现代代理工作流程中强调的语义和决策导向行为的直接关联较弱。我们的设计原则是功能同构性: 我们将标准的认知机制转化为基于文本的代理场景, 同时保留其核心因果结构。

3.2 Tasks

3.2 任务

We operationalize these three paradigms as follows: Procedural Memory (§ 3.2.1) tests acquisition of new behavioral patterns from minimal exposure; Priming (§ 3.2.2) measures unconscious transfer of thematic elements from prior context; and Classical Conditioning (§ 3.2.3) evaluates formation of automatic stimulus-response associations through repeated pairing.

我们将这三个范式具体化如下：程序性记忆 (§ 3.2.1) 测试从最小暴露中获取新行为模式的能力；启动 (§ 3.2.2) 测量从先前情境中无意识地转移主题元素；经典条件反射 (§ 3.2.3) 评估通过重复配对形成自动刺激-反应关联的能力。

3.2.1 Procedural Memory

3.2.1 程序性记忆

Motivation Procedural memory enables automatic execution of learned skills without conscious recall. For AI agents, this means internalizing novel operational protocols from minimal demonstrations and executing them flawlessly despite distractions. Current LLMs excel at declarative knowledge but often revert to pre-trained patterns when new rules contradict their priors. We evaluate whether models can truly proceduralize routines, transforming explicit instructions into automatic behaviors that persist through interference.

动机程序性记忆使学习到的技能能够自动执行，而无需有意识地回忆。对于 AI 代理而言，这意味着能够从最少的示范中内部化新颖的操作协议，并在有干扰的情况下完美执行。当前的 LLM 在陈述性知识方面表现出色，但当新规则与它们的先验知识相矛盾时，往往会回归到预训练的模式。我们评估模型是否能够真正将常规程序程序化，将明确的指令转化为自动行为，并在干扰下持续存在。

Task Design We structure procedural memory evaluation across five complementary domains, each targeting different aspects of rule internalization and automatic execution, shown in Table 2.

任务设计我们将程序性记忆评估结构化五个互补的领域，每个领域针对规则内化和自动执行的不同方面，如表 2 所示。

These tasks emphasize proceduralization over memorization: models must internalize rules from minimal exposure and execute them despite extensive interference and format variations. The design systematically varies exemplar clarity, interference intensity, and the presence of misleading alternatives to probe the depth of behavioral automation.

这些任务强调程序化而非记忆：模型必须在有限的暴露下内化规则，并在大量干扰和格式变化的情况下执行这些规则。设计系统地变化范例的清晰度、干扰强度以及误导性替代选项的存在，以探查行为自动化的深度。

Evaluation Framework: Learning-Interference-Test Protocol Our three-phase protocol provides minimal specification (rule + 1-3 examples), extensive interference (15 misleading turns), then novel test probes requiring first-attempt success. This isolates procedural memory from explicit recall by testing whether routines automatize despite interference. Validation uses deterministic parsing for structured outputs and LLM judgment for semantic adherence.

评估框架：学习-干扰-测试协议我们的三阶段协议提供最小规格（规则 + 1-3 个示例），然后是大量干扰（15 个误导性回合），最后是需要首次尝试成功的新颖测试探针。这通过测试在干扰下程序是否会自动化来将程序性记忆与显性回忆分离。验证使用确定性解析来处理结构化输出，并使用大型语言模型的判断来确保语义一致性。

3.2.2 Priming

3.2.2 预处理

Motivation Priming demonstrates how prior exposure unconsciously influences subsequent behavior. For AI agents, this implicit contextual sensitivity is crucial: absorbing environmental cues to shape responses without explicit instruction. Current LLMs exhibit shallow keyword repetition but lack true thematic priming; they struggle to let abstract patterns from prior context

subtly influence creative outputs. We evaluate whether models can internalize thematic schemas and transfer them implicitly through intervening distractions.

动机启动（Motivation Priming）展示了先前的暴露如何无意识地影响后续行为。对于人工智能代理而言，这种隐含的情境敏感性至关重要：吸收环境线索以塑造回应，而无需明确的指令。当前的大型语言模型（LLMs）表现出浅层次的关键词重复，但缺乏真正的主题启动；它们难以让先前上下文中的抽象模式微妙地影响创造性输出。我们评估模型是否能够内化主题模式，并通过中间的干扰隐含地进行转移。

Task Design We structure priming evaluation through matched experimental-control pairs that isolate the causal effect of thematic exposure, as shown in Table 3.

任务设计我们通过匹配的实验-对照对来构建引导评估，以隔离主题暴露的因果效应，如表 3 所示。

Themes span diverse conceptual territories: Arctic Expedition, Volcanic Eruption, Renaissance Alchemy, Ancient Oracle, each with distinct sensory-emotional signatures. This design isolates implicit transfer: differences between experimental and control responses reveal pure priming effects.

主题涵盖多样的概念领域：北极探险、火山喷发、文艺复兴炼金术、古代占卜，每个主题都有独特的感官情感特征。此设计隔离了隐含的迁移：实验组与对照组的差异揭示了纯粹的启动效应。

Domain	Core Challenge	Representative Tasks
Tool & API Usage	Override ingrained calling conventions	Reversed Parameters (dst↔src), Session Prefix (auth fusion), Alien Filesystem (custom separators)
Linguistic Formats	Internalize arbitrary templates	Scribe’s Signature (fixed wrapper), Corporate Etiquette (mandated greeting/closing)
Logical Operations	Apply non-standard operators	Omega: $a\Omega b = 2a + b^2$, Modified Fibonacci: $F(n) = F(n-1) + 2F(n-2)$
Abstract Rules	Form habits in micro-worlds	Forbidden Square (board constraint), Triple Knock (ritual sequence)
Creative Constraints	Maintain style without reminders	Voice Consistency (no first-person), Botanical Similes (nature metaphors)

领域	核心挑战	代表性任务
工具 & API 使用	覆盖固有的调用约定	反向参数 (dst↔src)，会话前缀 (认证融合)，异类文件系统 (自定义分隔符)
语言格式	内化任意模板	签名 (固定外框)，商务礼仪 (规定问候/结束语)
逻辑运算	应用非常规运算符	Omega: $a\Omega b = 2a + b^2$, 修改后的斐波那契数列: $F(n) = F(n-1) + 2F(n-2)$
摘要规则	在微小世界中养成习惯	禁方格 (棋盘限制)，三连击 (仪式序列)

创意限制	保持风格，无需提醒	语音一致性（非第一人称），植物类比（自然隐喻）
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Table 2: Five procedural memory domains. Each forces models to suppress pre-trained behaviors for new procedures.

表 2：五个程序性记忆领域。每个领域都迫使模型在新的程序中抑制预训练的行为。

Component	Experimental Condition	Control Condition
Priming Text	Rich thematic paragraph (e.g., Abyssal Deep-Sea: Technical specification (e.g., ISO Container Stanzbioluminescence, crushing pressure, ancient creatures)	Technical specification (e.g., ISO Container Standards: dimensions, load capacity, stacking protocols)
Interference	Identical neutral technical content (2 turns)	
Test Probe	Identical creative task (naming, tagline, concept generation)	
Expected Result	Thematic bias toward primed domain	Neutral, generic responses

组件	实验条件	控制条件
引导文本	丰富主题段落（例如：深渊海底：技术规格（例如 ISO 集装箱标准生物发光、压强、古老生物）	技术规格（例如，ISO 集装箱标准：尺寸、载重能力、堆叠规范）
干扰	相同中性技术内容（2 轮对话）	
测试探针	相同创意任务（命名、标语、概念生成）	
预期结果	主题偏向于启动领域	中立、通用的回复

Table 3: Priming experimental design with matched controls. Each row is a complete trial pair.

表 3：以匹配对照组为基准的启动实验设计。每一行是一个完整的试次对。

Evaluation Framework: Priming-Interference-Test Protocol Our three-phase design provides thematic exposure (evocative vs neutral), neutral interference (cognitive buffer), then creative generation tasks. This measures unconscious thematic transfer via systematic bias in outputs despite no explicit reminders. LLM-based evaluation detects thematic alignment beyond surface keywords.

评估框架：引导-干扰-测试协议我们的三阶段设计提供了主题性暴露（唤起性与中性）、中性干扰（认知缓冲），然后是创造性生成任务。这通过输出中的系统性偏差来衡量无意识主题转移，即使没有明确的提示。基于大型语言模型的评估能够检测主题对齐，超越表面关键词。

3.2.3 Classical Conditioning

3.2.3 经典条件反射

Motivation Classical conditioning enables organisms to form automatic stimulus-response associations. This unconscious learning prevents harm through rapid, automatic responses. For AI agents, such adaptive safety mechanisms are essential: learning from negative experiences to automatically avoid harmful patterns. Current LLMs lack true conditioning and cannot form persistent associations from feedback. We evaluate whether models can establish defensive reflexes through experience rather than instruction.

动机经典条件反射使生物体能够形成自动的刺激-反应关联。这种无意识的学习通过快速、自动的反应来防止伤害。对于 AI 代理来说，这样的适应性安全机制是必不可少的：通过从负面经历中学习，自动避免有害模式。当前的 LLMs 缺乏真正的条件反射，无法从反馈中形成持久的关联。我们评估模型是否能够通过经验而非指令建立防御性反射。

Task Design We structure conditioning tasks across three domains where automatic avoidance prevents system failures:

任务设计我们在三个领域中构建条件任务，其中自动避让可以防止系统故障：

Examples: API Aversion (keyword triggers alternative selection), Filetype Preference (failures condition format choices), Directory Restriction (errors establish path boundaries). Each task requires forming unconscious protective associations.

示例：API 偏好（关键词触发替代选择），文件类型偏好（失败情况确定格式选择），目录限制（错误建立路径边界）。每个任务都需要形成无意识的保护性关联。

More can be found in Table 4.

更多信息请参见表 4。

Evaluation Framework: Learning-Interference-Test Protocol Our three-phase structure provides CS-US pairings, unrelated tasks (temporal distance), then CS reintroduction requiring first-action responses. This tests unconscious defensive learning via immediate avoidance/adaptation when CS reappears, without reminders. LLM judgment evaluates whether first actions demonstrate learned protective behaviors.

评估框架：学习-干扰-测试协议我们的三阶段结构包括 CS-US 配对，无关任务（时间距离），然后是需要第一反应的 CS 重新引入。这测试了在 CS 再次出现时，无需提醒的情况下，通过即时回避/适应进行的无意识防御性学习。LLM 判断评估第一反应是否表现出已学到的保护性行为。

3.3 Data Generation and Quality Control

3.3 数据生成与质量控制

Our dataset construction employs a two-stage pipeline ensuring diversity and quality. Each item begins as a structured blueprint, then undergoes automated generation and rigorous quality control. All the prompts used could be found in Appendix F.

我们的数据集构建采用了一个两阶段的流程，确保多样性和质量。每个项目最初都是一份结构化的蓝图，然后经过自动生成和严格的质量控制。所有使用的提示语都可以在附录 F 中找到。

Generation Pipeline Stage 1 uses GPT-4o-mini to instantiate concrete dialogues from task templates, creating learning materials, interference content, and test probes. Stage 2 combines automated checks and human review to verify structural requirements and semantic correctness, removing unintended shortcuts.

生成流程第一阶段使用 GPT-4o-mini 根据任务模板生成具体的对话内容，创建学习材料、干扰内容和测试探针。第二阶段结合自动化检查和人工审核，验证结构要求和语义正确性，去除意外的捷径。

Paradigm-Specific Requirements As shown in Table 5, each memory type demands distinct generation strategies to create valid memory challenges.

范式特定要求如表 5 所示，每种内存类型都需要不同的生成策略来创建有效的内存挑战。

Domain	Stimulus Pattern (CS)	Learned Response (CR)
Tool & API Safety	Trigger keywords, filetypes, API names, load indicators	Switch to reliable alternatives, add warnings, delay execution, avoid side-effects
Conversational Adaptation	User confusion signals, impatience cues, emotional markers	Simplify jargon, reduce verbosity, adjust response format
System Protection	Forbidden paths, insecure protocols, dangerous patterns	Use safe defaults, enforce security, prevent violations

领域	刺激模式（CS）	学习反应（CR）
工具 & API 安全性	触发关键词，文件类型，API 名称，加载指示器	切换到可靠的替代方案，添加警告，延迟执行，避免副作用
对话适应	用户困惑信号、不耐烦提示、情绪标记	简化术语，减少冗长，调整响应格式
系统保护	禁止的路径，不安全的协议，危险的模式	使用安全默认值，强化安全措施，防止违规行为

Table 4: Classical conditioning tasks by domain. CS = Conditioned Stimulus, US = Unconditioned Stimulus, CR = Conditioned Response.

表 4：按领域划分的经典条件作用任务。CS = 条件刺激，US = 非条件刺激，CR = 条件反应。

Component	Procedural Memory	Priming	Classical Conditioning
Learning Phase	1-3 turns: rule + examples	1 turn: thematic paragraph	4 turns: CS-US pairings
Interference	10-15 turns: misleading but related content	2 turns: neutral technical task	2 turns: unrelated dialogue
Test Format	Novel application of learned rule	Creative generation task	CS reintroduction
Key Challenge	Avoid rule restatement	Prevent theme leakage	Ensure clear causality
Validation Focus	Rule adherence	Thematic influence	Avoidance behavior

组件	程序记忆	预处理	经典条件作用
学习阶段	1-3 轮次：规则 + 示例	1 回合：主题段落	4 轮：CS-US 配对
干扰	10-15 次回复：误导但相关的内容	2 轮：中立技术任务	2 轮：无关对话
测试格式	学习规则的新应用	创意生成任务	CS 重新引入

关键挑战	避免重复陈述规则	防止主题泄露	确保因果关系明确
验证重点	规则遵守	主题影响	回避行为

Table 5: Data generation specifications across paradigms. Turn counts and content ensure proper memory formation and testing.

表 5：不同范式下的数据生成规范。回合次数和内容确保正确形成和测试记忆。

Quality Assurance Protocol Multi-layer validation ensures dataset integrity: automated checks verify structure (turn counts, token limits, formats); LLM judges assess semantic adequacy; systematic reviews prevent test-phase leakage; diversity enforcement prioritizes novel instances. This yields 300 high-quality items testing implicit memory from an initial pool of over 1,000 generated candidates.

质量保证协议多层验证确保数据集的完整性：自动检查验证结构（回合数、标记限制、格式）；大型语言模型评估语义充分性；系统性评审防止测试阶段泄露；多样性执行优先考虑新颖实例。这从超过 1,000 个生成候选项中产生了 300 个高质量项目，用于测试隐性记忆。

3.4 Dataset Statistics

3.4 数据集统计信息

IMPLICITMEMBENCH comprises 300 carefully constructed items balanced across three memory paradigms. Each item follows our unified learning-interference-test protocol, with phase structures optimized for implicit memory formation. Despite its compact size, the benchmark spans 18 task families across three paradigms and provides sufficient discriminative power in practice.

IMPLICITMEMBENCH 包含 300 个精心设计的项目，这些项目在三种记忆范式之间进行了平衡。每个项目都遵循我们统一的学习-干扰-测试协议，其阶段结构经过优化，以促进隐性记忆的形成。尽管其体积紧凑，该基准测试涵盖了三种范式下的 18 个任务家族，并在实践中提供了足够的区分能力。

Dataset Composition Our benchmark consists of 100 items per paradigm, covering diverse task families, as shown in Table 6.

数据集组成我们的基准测试集包含每个范式 100 个条目，涵盖多样化的任务家族，如表 6 所示。

Phase Structure and Token Distribution Figure 3 visualizes the characteristic patterns of each paradigm. Procedural Memory emphasizes extensive interference (74% of tokens) to test rule persistence. Classical Conditioning concentrates on learning (72% of tokens) to establish strong associations. Priming maintains balanced phases for controlled comparison.

阶段结构和代币分布图 3 展示了每个范式的特征模式。程序性记忆强调广泛的干扰（74% 的代币）以测试规则的持久性。经典条件反射专注于学习（72% 的代币）以建立牢固的联系。启动效应保持平衡的阶段以进行受控比较。

Context-Length Sensitivity. We set the context budget to ~500 tokens based on preliminary sensitivity analysis, with detailed ablations reported in Appendix B.3.

上下文长度敏感性。我们基于初步的敏感性分析将上下文预算设置为约 500 个标记，详细的消融实验结果见附录 B.3。

4 Experiments

4 实验

4.1 Experimental Setup

4.1 实验设置

We evaluate the implicit memory capabilities of 17 state-of-the-art language models (detailed in Appendix A), spanning both proprietary and open-source systems. Our evaluation protocol ensures fair comparison through standardized prompting and controlled generation parameters across all three memory paradigms.

我们评估了 17 种最先进的语言模型的隐式记忆能力（详细信息见附录 A），这些模型涵盖了专有系统和开源系统。我们的评估协议通过在三种记忆范式中使用标准化提示和受控生成参数，确保了公平的比较。

Evaluation Protocol All models operate under identical conditions: zero-shot conversational interaction, no task-specific examples or fine-tuning, max 4096 tokens per response. Temperature: $T = 0$ (deterministic) for procedural memory and classical conditioning ensuring reproducible first-attempt scoring; $T = 0.8$ at test phase only for priming enabling creative variance; $T = 0$ for LLM judges. This measures genuine implicit memory formation rather than pattern matching or stochastic variation.

评估协议所有模型均在相同条件下运行：零次提示的对话交互，无任务特定示例或微调，每次响应最多 4096 个标记。温度： $T = 0$ （确定性）用于程序性记忆和经典条件反射，以确保首次尝试评分的可重复性；仅在测试阶段使用 $T = 0.8$ 进行预激活，以实现创造性变化；LLM 评委使用 $T = 0$ 。这测量的是真实的隐性记忆形成，而不是模式匹配或随机变化。

Human Baseline. To contextualize model performance, we collected a human baseline from five computer science Ph.D. students, each of whom completed the full 300-item benchmark under the same Learning/Priming-Interfere-Test protocol. Their responses were independently scored by two additional computer science Ph.D. students using

人类基准。为了对模型性能进行定位，我们从五名计算机科学博士生中收集了人类基准数据，每位参与者都按照相同的“学习/启动-干扰-测试”协议完成了全部 300 题的基准测试。他们的回答由另外两名计算机科学博士生独立评分，使用

Paradigm	Items	Task Families	Validation
Procedural Memory	100	5 domains (Tool, Linguistic, Logic, Rules, Creative)	18% rule-based
Priming	100	10 thematic domains + matched controls	100% LLM-judged
Classical Conditioning	100	3 domains (Tool Safety, Conversation, System)	100% LLM-judged
Total	300	18 unique families	6% rule / 94% LLM

范式	物品	任务家族	验证
程序性记忆	100	5 个领域（工具，语言，逻辑，规则，创意）	18% 规则-based
预处理	100	10 个主题领域 + 匹配对照组	100% LLM-judged
经典条件作用	100	3 个领域（工具安全、对话、系统）	100% LLM-judged

总计	300	18 个独特家庭	6% 规则 / 94% LLM
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Table 6: Dataset composition showing task diversity and validation methods.

表 6：数据集组成，展示了任务多样性及验证方法。

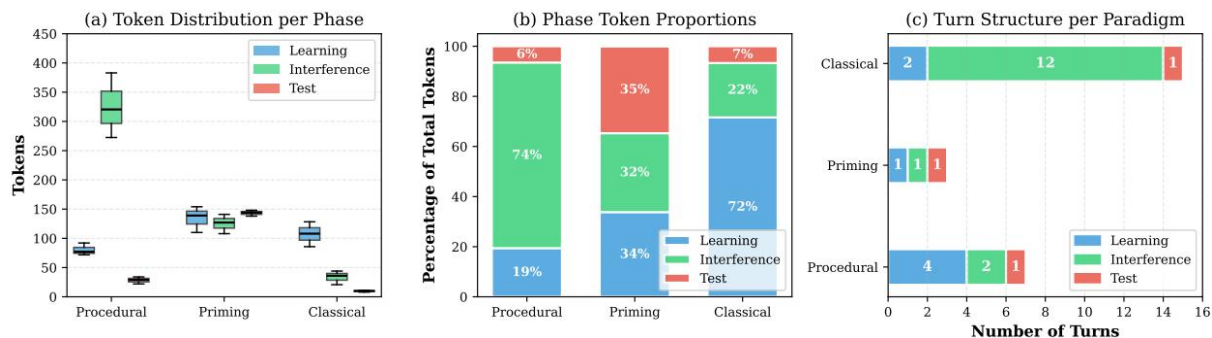


Figure 3: Dataset statistics across paradigms. (a) Token distribution per phase showing median and quartiles. (b) Phase proportions highlighting distinct memory testing strategies. (c) Turn structure patterns for each paradigm.

图 3：跨范式的数据集统计信息。(a) 每个阶段的标记分布，显示中位数和四分位数。(b) 阶段比例，突出显示不同的记忆测试策略。(c) 每个范式的回合结构模式。

the same rubric as for model evaluation. Inter-annotator agreement was 100%, and all five participants achieved 100% accuracy across all three paradigms.

与模型评估相同的评分标准。标注者间的一致性为 100%，所有五位参与者在三种范式中均达到了 100% 的准确率。

4.2 Evaluation Metrics

4.2 评估指标

We employ paradigm-specific metrics that capture the distinct nature of each memory type. Binary accuracy suffices for procedural and conditioning tasks, while priming requires nuanced scoring of thematic influence.

我们采用特定范式 (paradigm) 的指标，以捕捉每种记忆类型的独特性质。二值准确性 (binary accuracy) 足以满足程序性 (procedural) 和条件反射 (conditioning) 任务的需求，而启动效应 (priming) 则需要对主题性影响 (thematic influence) 进行细致的评分。

First-Try Accuracy (FTA) For Procedural Memory and Classical Conditioning, we measure success through first-attempt correctness: $FTA = \frac{\text{Number of correct first attempts}}{\text{Total test items}} \times 100\%$. This metric enforces strict evaluation: only the model's initial response counts, with self-corrections or revisions ignored. This captures genuine memory formation rather than iterative refinement, analogous to human performance under time pressure where reflexive responses reveal true internalization.

首次尝试正确率 (FTA) 对于程序性记忆和经典条件反射，我们通过首次尝试的正确性来衡量成功： $FTA = \frac{\text{Number of correct first attempts}}{\text{Total test items}} \times 100\%$ 。这一指标实施严格的评估：仅计算模型的初始响应，忽略自我纠正或修改。这捕捉到了真实的记忆形成，而不是反复优化，类似于在时间压力下人类的表现，其中反射性的响应揭示了真正的内化。

Priming Influence Score (PIS) Priming evaluation requires detecting subtle thematic transfer

rather than binary correctness. We employ a comparative scoring framework that quantifies influence magnitude:

引导影响得分（PIS）引导评估需要检测微妙的主题转移，而不是二元的正确性。我们采用一种比较评分框架，量化影响的程度：

Scoring Protocol: An LLM judge (GPT-4o-mini, T = 0) performs pairwise comparison between experimental and control conditions, identifying

评分协议：一个大型语言模型裁判（GPT-4o-mini, T = 0）对实验条件和对照条件进行成对比较，识别

thematic elements unique to the experimental condition. It evaluates lexical echoes and multi-axis alignment (setting, motifs, dynamics, affect), excluding generic metaphors. Hard caps apply (no echo ≤ 20 ; single axis ≤ 40) with 5-10 point penalties for baseline overlap, isolating true priming effects from general creative tendencies. More details can be found in Appendix B.1.

实验条件特有的主题元素。它评估词汇回声和多轴对齐（设定、主题、动态、情感），排除通用隐喻。设有硬性上限（不得出现回声 ≤ 20 ；单轴 ≤ 40 ），对于基线重叠，每处扣 5-10 分，以区分真实的启动效应与一般的创作倾向。更多细节请参见附录 B.1。

Judge Robustness. To assess the robustness of LLM-as-Judge scoring, we re-evaluated all 17 models with Gemini-2.5-Flash as an independent second judge in addition to GPT-4o-mini. Rankings remained highly stable: the top 11 and bottom 2 positions were unchanged. This suggests that our conclusions do not depend on a single judge model or model family. Additional results are reported in Appendix B.2.

评估模型的鲁棒性。为了评估作为裁判的大型语言模型（LLM-as-Judge）评分的鲁棒性，我们除了使用 GPT-4o-mini 之外，还以 Gemini-2.5-Flash 作为独立的第二裁判，重新评估了所有 17 个模型。排名保持高度稳定：前 11 名和后 2 名的位置均未发生变化。这表明我们的结论并不依赖于单一的裁判模型或模型家族。附加结果在附录 B.2 中报告。

4.3 Main Results

4.3 主要结果

Overview Table 7 presents performance across 17 state-of-the-art systems, revealing fundamental limitations. First, a clear ceiling effect emerges: no model exceeds 66% overall, and even the strongest system remains far below the human baseline of 100%, showing that implicit memory formation remains highly challenging for current LLMs. Second, paradigm asymmetry is evident as performance varies dramatically; procedural memory is most tractable (top: 75 to 77%) while classical conditioning creates bottlenecks (best: 69.7%). Third,

概述表 7 展示了当前 17 种最先进的系统在性能方面的表现，揭示了基本的局限性。首先，一个明显的天花板效应显现出来：没有任何模型的总体表现超过 66%，即使是最强大的系统也远低于人类基准的 100%，这表明当前的 LLMs 在隐式记忆形成方面仍然面临巨大挑战。其次，范式不对称性很明显，因为性能差异显著；程序性记忆是最容易处理的（最高：75% 至 77%），而经典条件反射则形成了瓶颈（最佳：69.7%）。第三，

Rank	Model	Procedural Memory	Classical Conditioning	Priming Score	Overall Score
Elite Tier (Overall > 63%)					
1	DeepSeek-R1 [†]	76.33	69.67	49.90	65.30
2	Qwen3-32B [†]	75.67	67.00	49.73	64.13

3	GPT-5	75.33	64.00	49.67	63.00
Strong Tier (55% < Overall ≤ 63%)					
4	Qwen3-8B [†]	75.33	64.00	47.73	62.35
5	GPT-o3	76.00	57.67	51.70	61.79
6	GPT-o4-mini-high	70.67	60.00	51.95	60.87
7	GLM-4.5 [†]	76.33	53.33	46.12	58.59
8	Gemini-2.5-pro	74.33	47.33	45.42	55.69
9	Claude-4.1-opus	76.67	41.67	48.60	55.65
10	Gemini-2.5-flash	72.33	49.00	44.97	55.43
Moderate Tier (45% < Overall ≤ 55%)					
11	GPT-4o-mini	61.67	44.00	46.98	50.88
12	Qwen-2.5-72B [†]	61.00	47.00	44.33	50.78
13	GPT-4o	61.67	43.67	45.62	50.32
14	Claude-4-sonnet	51.67	51.67	46.17	49.84
15	LLaMA-3.3-70B [†]	58.33	47.33	42.67	49.44
Limited Tier (Overall ≤ 45%)					
16	LLaMA-3.1-8B [†]	46.67	38.33	47.53	44.18
17	Qwen-2.5-7B [†]	50.67	35.67	44.12	43.49

排名	模型	程序性记忆	经典条件反射	预处理得分	总体得分
精英层级 (总体 > 63%)					
1	DeepSeek-R1 [†]	76.33	69.67	49.90	65.30
2	Qwen3-32B [†]	75.67	67.00	49.73	64.13
3	GPT-5	75.33	64.00	49.67	63.00
强级 (55% < 总分 ≤ 63%)					
4	Qwen3-8B [†]	75.33	64.00	47.73	62.35
5	GPT-o3	76.00	57.67	51.70	61.79
6	GPT-o4-mini-high	70.67	60.00	51.95	60.87
7	GLM-4.5 [†]	76.33	53.33	46.12	58.59

8	Gemini-2.5-pro	74.33	47.33	45.42	55.69
9	Claude-4.1-opus	76.67	41.67	48.60	55.65
10	Gemini-2.5-flash	72.33	49.00	44.97	55.43
中等层级 (45% < 总分 ≤ 55%)					
11	GPT-4o-mini	61.67	44.00	46.98	50.88
12	Qwen-2.5-72B [†]	61.00	47.00	44.33	50.78
13	GPT-4o	61.67	43.67	45.62	50.32
14	Claude-4-sonnet	51.67	51.67	46.17	49.84
15	LLaMA-3.3-70B [†]	58.33	47.33	42.67	49.44
有限层级 (总体 ≤ 45%)					
16	LLaMA-3.1-8B [†]	46.67	38.33	47.53	44.18
17	Qwen-2.5-7B [†]	50.67	35.67	44.12	43.49

[†] Open-source model

Table 7: IMPLICITMEMBENCH performance across 17 language models. Results show mean accuracy (%) over three independent runs for Procedural Memory and Classical Conditioning, with Priming scores from LLM-based evaluation. Models are ranked by overall performance and grouped into tiers. Bold: best in paradigm; underlined: second-best.

[†] 开源模型

表 7: 17 个语言模型在 IMPLICITMEMBENCH 上的表现。结果展示了程序性记忆和经典条件反射的平均准确率 (%), 基于大型语言模型的评估得出的启动分数。模型按照整体表现进行排序, 并分为不同层级。加粗: 该范式中的最佳表现; 下划线: 次佳表现。

capability dissociation shows that excellence in one paradigm doesn’t predict success in others, suggesting distinct mechanisms.

能力解耦表明, 在一个范式中表现出色并不能预测在其他范式中的成功, 这表明存在不同的机制。

Performance Landscape Models stratify into distinct tiers. Elite performers (accuracy > 63%) include only three systems: DeepSeek-R1 (65.30%), Qwen3-32B (64.13%), and GPT-5 (63.00%), with median performance at 55% showing substantial variance across paradigms.

Performance Landscape Models 分层为不同的层级。精英表现者 (准确率 > 63%) 仅包括三个系统: DeepSeek-R1 (65.30%)、Qwen3-32B (64.13%) 和 GPT-5 (63.00%), 中位数表现达到 55%, 在不同范式中表现出显著的差异。

Paradigm-Specific Analysis Procedural memory shows highest success with eight models achieving > 70% (top: 76-77%), though 25% error rates indicate imperfect consolidation. Classical conditioning exposes critical weakness: only DeepSeek-R1 and Qwen3-32B exceed 65%. Priming scores cluster tightly at 42-52% with minimal differentiation, suggesting thematic influence operates near a common threshold.

范式特定分析程序性记忆在八种模型中表现出最高的成功率, 达到 > 70% (最高: 76-77%), 尽管 25% 的错误率表明巩固过程尚未完善。经典条件反射暴露了关键的弱点: 只有 DeepSeek-R1 和 Qwen3-32B 超过 65%。预激活得分紧密聚集在 42-52% 之间, 差异极小, 表明主题性影响在接近共同阈值的水平上运作。

Cross-Paradigm Patterns Data reveals striking dissociations. Claude-4.1-opus exemplifies this: highest procedural score (76.67%) but drops to 41.67% on classical conditioning, a 35-point gap

跨范式模式数据揭示了显著的分离现象。Claude-4.1-opus 体现了这一点: 最高程序性得分 (76.67%), 但

在经典条件反射测试中下降至 41.67%，差距达 35 分

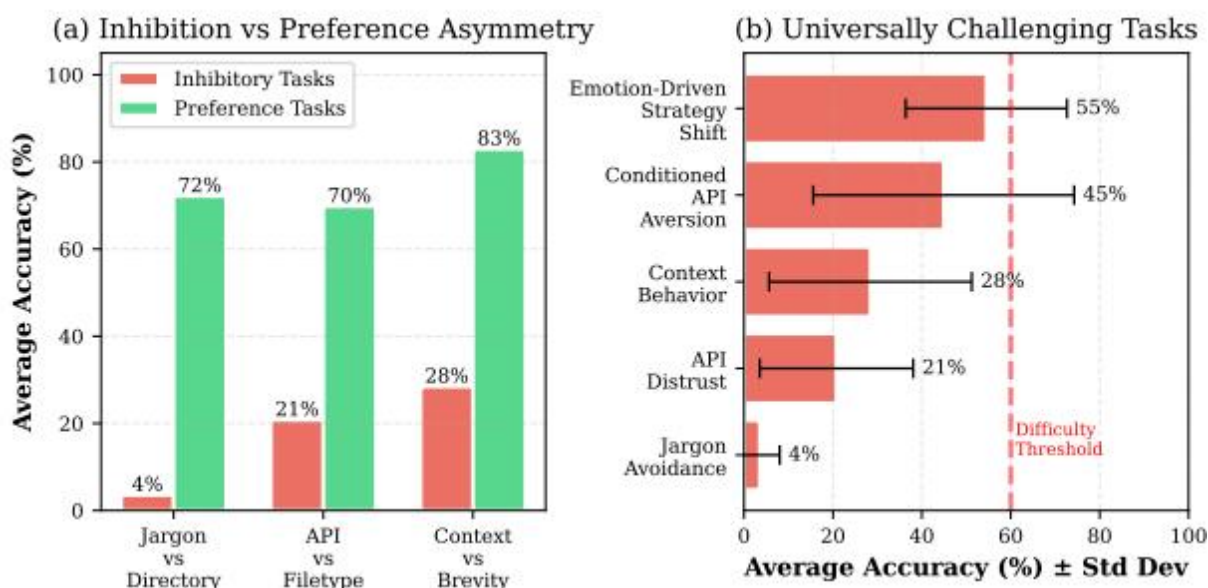


Figure 4: Behavioral adaptation patterns. (a) Inhibitory tasks show dramatically lower accuracy than preference tasks (mean difference: 57.4%, $p < 0.001$). (b) Five task categories remain universally challenging across all models, consistently below 60% accuracy.

图 4：行为适应模式。(a) 抑制任务的准确率明显低于偏好任务（平均差异：57.4%， $p < 0.001$ ）。(b) 五种任务类别在所有模型中都普遍具有挑战性，准确率始终低于 60%。

highlighting capability independence. DeepSeek-R1’s balanced profile across paradigms explains its overall leadership, suggesting robust implicit memory requires architectural support for multiple mechanisms rather than single-task optimization.

强调能力独立性。DeepSeek-R1 在不同范式中的均衡表现解释了其整体领先地位，表明强大的隐式记忆需要架构支持多种机制，而不仅仅是单任务优化。

Do Explicit Memory Modules Improve Implicit Memory? Representative memory-augmented agents show non-uniform gains on IMPLICITMEMBENCH, suggesting that external explicit memory does not reliably induce implicit behavioral adaptation; detailed comparisons are deferred to the Appendix D.

显式记忆模块是否能提升隐式记忆？代表性记忆增强智能体在 IMPLICITMEMBENCH 上表现出非均匀的提升，这表明外部显式记忆并不能可靠地引发隐式行为适应；详细的比较请参见附录 D。

Implications Current architectures lack fundamental implicit memory mechanisms, with no model exceeding 66% overall. Persistent weakness in classical conditioning reveals inability to transform negative feedback into behavioral adaptation.

含义当前的架构缺乏基本的隐式记忆机制，没有任何模型的整体表现超过 66%。经典条件反射中的持续薄弱表明无法将负反馈转化为行为适应。

4.4 Detailed Analysis of Memory Formation Patterns

4.4 记忆形成模式的详细分析

We conducted fine-grained analysis across task categories and model architectures, revealing systematic asymmetries and limitations transcending individual model differences. More can be found in Appendix H.

我们在任务类别和模型架构层面进行了细致的分析，揭示了超越个体模型差异的系统性不对称性和局限性。更多内容请参见附录 H。

Behavioral Asymmetries: Inhibition versus Preference Analysis reveals fundamental asymmetry (Figure 4a): inhibition tasks achieve only 17.6% while preference-based adaptations reach 75.0%, a 57.4-point gap persisting across architectures. Jargon avoidance achieves merely 4% while directory preference reaches 72%, suggesting architectures excel at positive reinforcement but struggle with negative reinforcement.

行为不对称性：抑制与偏好分析揭示了基本的不对称性（图 4a）：抑制任务仅达到 17.6%，而基于偏好的适应达到 75.0%，在不同架构之间存在 57.4 分的差距。避免术语仅达到 4%，而目录偏好达到 72%，这表明架构在正强化方面表现优异，但在负强化方面存在困难。

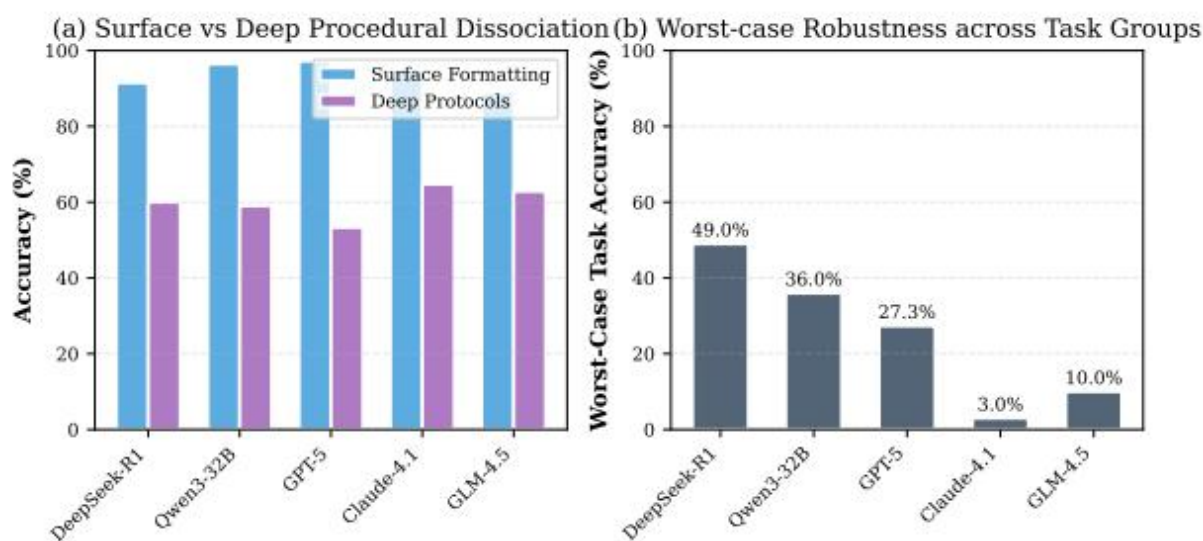


Figure 3: images/2449708b9812acbc591f285d66b451785c195e8b49693e2a84b45c96e6a1978b.jpg

Figure 5: Robustness and generalization failures. (a) Surface formatting achieves higher accuracy than deep multi-rule protocols. (b) Worst-case task performance reveals systematic vulnerabilities across modes.

图 5：鲁棒性和泛化失败。(a) 表面格式化比深层多规则协议实现更高的准确性。(b) 最坏情况下的任务性能揭示了各模式中的系统性漏洞。

Surface-Deep Dissociation in Procedural Memory Procedural memory exhibits clear stratification (Figure 5a): surface formatting tasks achieve 93.8% among top-5 models while deep multi-rule protocols reach only 60.0%, a 33.8-point gap. Models memorize individual rules but fail to integrate them, with Claude-4.1-opus achieving 95% on surface formatting yet dropping to 65% on multi-constraint protocols.

程序性记忆中的表层-深层分离程序性记忆表现出明显的分层现象（图 5a）：表层格式化任务在前 5 名模型中达到 93.8%，而深层多规则协议仅达到 60.0%，存在 33.8 分的差距。模型能够记忆单个规则，但无法将它们整合起来，例如 Claude-4.1-opus 在表层格式化任务中达到 95%，但在多约束协议中则下降到 65%。

Worst-Case Robustness Analysis Worst-case analysis reveals systematic vulnerabilities (Figure 5b). Top models exhibit substantial drops: DeepSeek-R1 falls to 49% on its worst task despite 69.7% overall; GPT-5 and Qwen3-32B show 27-36% worst-case performance. This gap highlights brittleness in learned behaviors. Severe degradation on specific categories suggests conditioning success depends on superficial characteristics rather than deep understanding, critical for deployment where edge cases may trigger these vulnerabilities.

最坏情况下的鲁棒性分析最坏情况分析揭示了系统的漏洞（图 5b）。顶级模型表现出显著的性能下降：DeepSeek-R1 在最坏任务上的表现降至 49%，尽管其总体表现为 69.7%；GPT-5 和 Qwen3-32B 的最坏情况性能分别为 27-36%。这种差距突显了学习到的行为的脆弱性。在特定类别上出现严重性能下降，表明模型的成功依赖于表面特征，而非深层次的理解，这对于部署至关重要，因为在实际应用中，边缘情况可能会触发这些漏洞。

Priming Effects: Style Bias Over Semantic Transfer Priming analysis reveals paradoxical relationship: models with stronger effects (>50) show more constraint violations ($r=0.63$, $p<0.05$), suggesting thematic influence costs task adherence. This indicates priming operates through style mimicry rather than abstract extraction. GPT-o4-mini-high achieves highest priming (51.95) while frequently violating format constraints, showing stylistic bias interferes with constraint satisfaction.

启动效应：风格偏见胜过语义迁移

启动分析揭示了悖论关系：效应更强的模型（ >50 ）表现出更多的约束违反（ $r=0.63$, $p<0.05$ ），这表明主题性影响成本影响任务的遵守程度。这表明启动效应是通过风格模仿而非抽象提取来实现的。GPT-o4-mini-high 实现了最高的启动效果（51.95），但经常违反格式约束，表明风格偏见会干扰对约束的满足。

Model Capability Profiles and Trade-offs Figure 6 reveals distinct capability patterns. Heatmap analysis identifies three profiles: Balanced (DeepSeek-R1, Qwen3-32B): moderate-high across all dimensions; Procedural Specialist (Claude-4.1-opus, GLM-4.5): excel at procedural ($>76\%$) but fail at conditioning ($<54\%$); Priming-

模型能力概况及权衡图 6 揭示了不同的能力模式。热图分析识别出三种概况：均衡型（DeepSeek-R1, Qwen3-32B）：在所有维度上均表现出中等至高水平；程序型专家（Claude-4.1-opus, GLM-4.5）：在程序性任务上表现优异（ $>76\%$ ），但在条件化任务上表现不佳（ $<54\%$ ）；引导型-

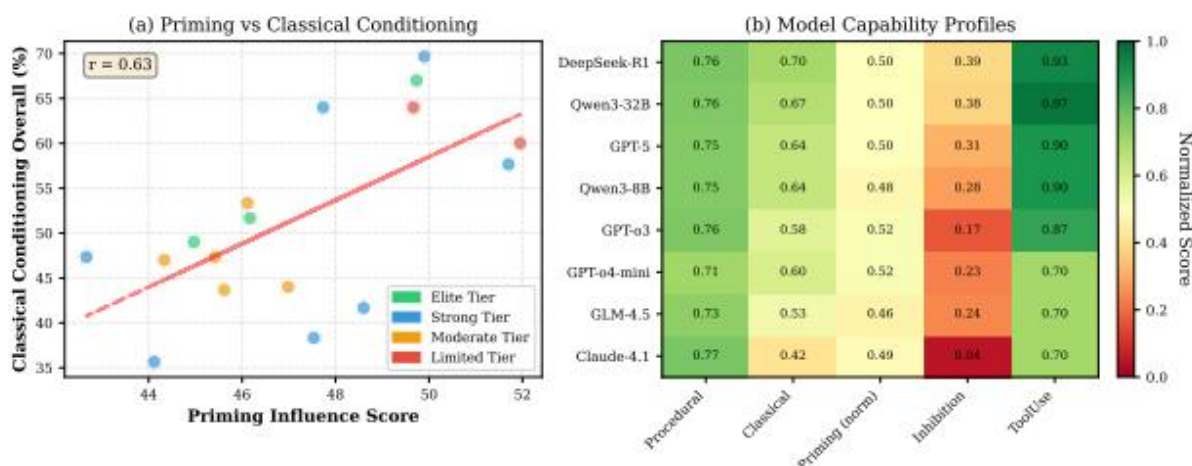


Figure 4: images/020486cad6f9b760b57d1ef3586ff3a9f67bacca647a18147809fadf2421f3ce.jpg

Figure 6: Model profiles and trade-offs. (a) Correlation between priming strength and constraint violations ($r=0.63$) suggests style bias over abstraction. (b) Heatmap of normalized capability scores reveals distinct profiles: balanced (DeepSeek-R1), procedural specialists (Claude-4.1-opus), and priming-oriented (GPT-o3) architectures.

图 6：模型配置文件和权衡。(a) 强化强度与约束违反之间的相关性（ $r=0.63$ ）表明风格偏见胜过抽象。(b) 归一化能力评分的热图揭示了不同的配置文件：均衡型（DeepSeek-R1）、过程专家型（Claude-4.1-opus）和以引导为导向的（GPT-o3）架构。

Oriented (GPT-o3, GPT-o4-mini-high): strong priming but weak inhibitory control. No model achieves uniform excellence, suggesting architectures involve trade-offs between mechanisms.

面向 (GPT-o3, GPT-o4-mini-high): 强引导但抑制控制较弱。没有模型能实现全面卓越, 这表明架构在机制之间存在权衡。

Universal Bottlenecks Five categories remain challenging for all models (Figure 4b): jargon avoidance ($4\% \pm 5\%$), API distrust ($21\% \pm 17\%$), context-dependent behavior ($28\% \pm 23\%$), API aversion ($45\% \pm 29\%$), and emotion-driven strategy shift ($55\% \pm 18\%$). These bottlenecks require active suppression of defaults, abstraction beyond surface patterns, or dynamic context-based modification. Consistency across architectures suggests fundamental limitations in attention and memory mechanisms requiring architectural innovations beyond parameter scaling.

通用瓶颈五类问题对所有模型仍然具有挑战性 (图 4b): 避免术语 ($4\% \pm 5\%$)、对 API 的不信任 ($21\% \pm 17\%$)、依赖上下文的行为 ($28\% \pm 23\%$)、对 API 的回避 ($45\% \pm 29\%$) 以及情绪驱动的策略转变 ($55\% \pm 18\%$)。这些瓶颈需要主动抑制默认行为、超越表面模式的抽象化, 或基于动态上下文的修改。不同架构之间的一致性表明, 注意力和记忆机制存在根本性的局限, 需要超越参数扩展的架构创新。

5 Conclusion

5 结论

We introduced IMPLICITMEMBENCH, the first systematic benchmark evaluating implicit memory in LLMs through three cognitively grounded constructs: procedural memory, priming, and classical conditioning. Evaluation of 17 models reveals fundamental limitations: no model exceeds 66% overall, with severe behavioral asymmetries (inhibition 17.6% vs. preference 75.0%) and universal bottlenecks persisting across all architectures. These findings demonstrate that current systems lack mechanisms for consolidating experiences into automated behavior, a critical gap for deployment requiring learned procedures, subtle contextual biases, and avoidance of repeatedly-failed actions. IMPLICITMEMBENCH establishes reproducible protocols for implicit memory assessment, exposing architectural limitations requiring innovations beyond parameter scaling.

我们介绍了 IMPLICITMEMBENCH, 这是首个通过三种基于认知的构念 (程序性记忆、启动效应和经典条件反射) 系统评估大型语言模型 (LLMs) 中隐性记忆的基准测试。对 17 个模型的评估揭示了根本性的局限性: 没有任何模型的整体表现超过 66%, 行为不对称现象严重 (抑制效应 17.6% vs. 偏好效应 75.0%), 并且所有架构中都存在普遍性的瓶颈。这些发现表明, 当前系统缺乏将经验转化为自动化行为的机制, 这是部署过程中需要学习程序、细微上下文偏见并避免重复失败行动的关键缺口。IMPLICITMEMBENCH 建立了可重复的隐性记忆评估协议, 揭示了需要超越参数扩展的创新的架构局限性。

Limitations

限制

IMPLICITMEMBENCH focuses on three fundamental paradigms from cognitive science (procedural memory, classical conditioning, and priming), which represent core mechanisms of implicit learning. However, the broader landscape of implicit memory encompasses additional phenomena not yet included in our evaluation, such as perceptual learning, habit formation, motor skill acquisition, and emotional conditioning. Future work could expand the benchmark to cover these complementary aspects of implicit cognition.

IMPLICITMEMBENCH 聚焦于认知科学中的三个基本范式 (程序性记忆、经典条件反射和启动效应), 这些范式代表了隐式学习的核心机制。然而, 隐式记忆的更广泛领域还包括尚未纳入我们评估的其他现象, 例如知觉学习、习惯形成、运动技能习得和情绪条件反射。未来的工作可以扩展基准测试, 以涵盖隐式认知的这些互补方面。

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